Microsoft Innovation Challenge

NextStep Science: A Lab AI Assistant

Team Members:

- Giovanni Yisrael Lopez Garcia
- Hansell Augusto Najarro Altan
- Isaac Suarez Maldonado
- Jose Angel Zavaleta Ruiz
- Sabina Ixchel Garcia Mendoza

I

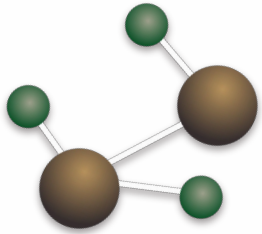
The Problem & Project Goal

The Problem & Project Goal

The Issue
Generic AI systems lack domain-aware safeguards and can provide hazardous scientific suggestions.

Value Proposition
Augmenting reasoning while preventing unsafe or hazardous recommendations.

The Goal
Build an assistant that understands experimental inputs (PDFs, CSVs, Images) to suggest **safe, bounded next steps.**

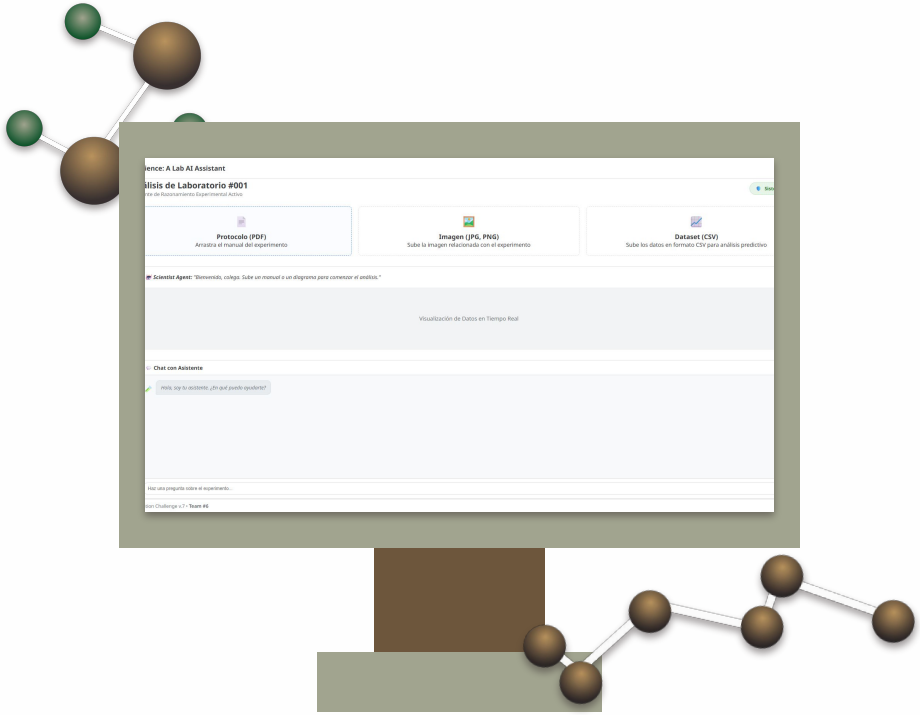


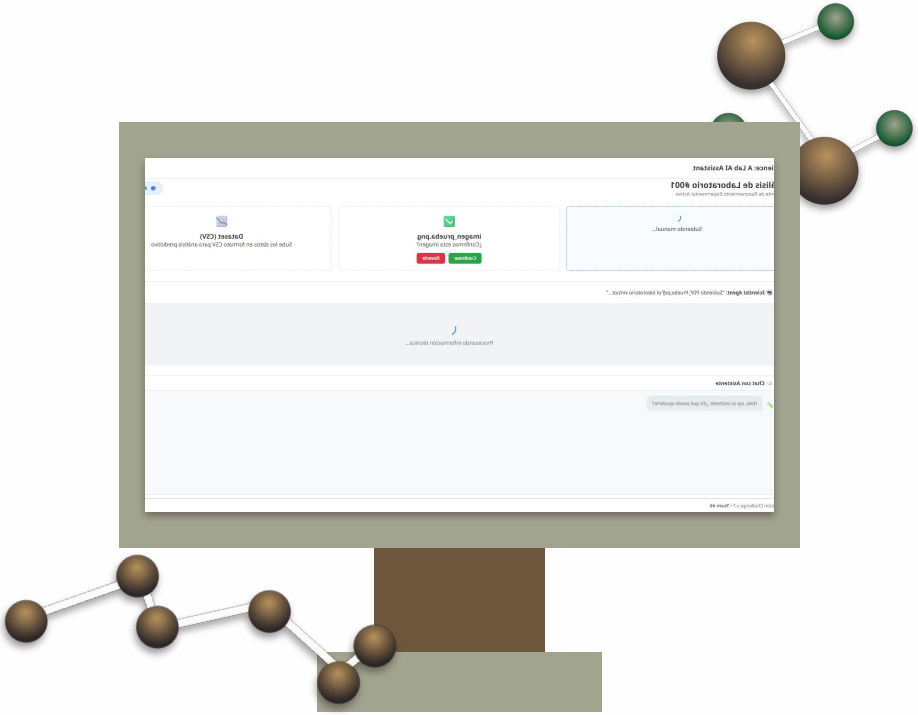
II

Core Capabilities & Multimodal Input

Input Handling

Supports **PDF** protocols, **CSV** datasets, and **image-based** experimental setups.





Processing Power

Automated extraction of experimental content and maintenance of conversational session memory.

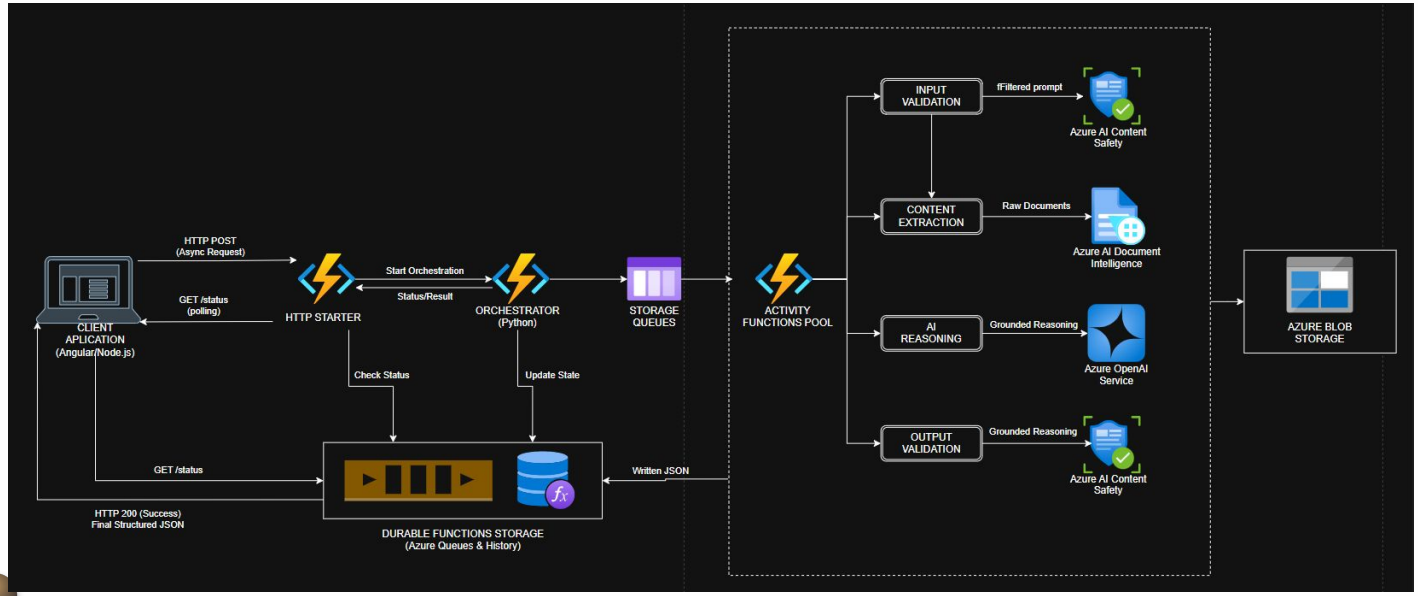
Outputs

Generates structured scientific **insights**, including **protocol analysis** and **safety assessments**

III

System Architecture (The "Durable" Engine)

System Architecture



Asynchronous Processing: Powered by Azure Durable Functions (Python) using an Orchestrator/Activity pattern

Storage: Secure integration with Azure Blob Storage for file persistence.
Communication: Reliable state management using Storage Queues and Durable Functions Storage.

IV

The AI Reasoning Layer

The AI Reasoning Layer

Model

Powered by **GPT-4.1** Mini via Azure AI Projects for cost-efficient and fast execution.

Agentic Intelligence

Uses an Azure AI Agent to generate comprehensive summaries, observations, and logical next steps.

Document Intelligence

Leverages Azure AI to extract text from complex scientific documents and setup photos.

V

Multi-Layer Safety (The "Golden Record")

Multi-Layer Safety

Layer 1 (Content Safety)

Input and output moderation via **Azure AI Content Safety.**

Layer 2 (Backend Logic)

Custom lab safety rules to block hazardous modifications and high-reactivity suggestions.

Layer 3 (Frontend Guard)

UI-level blocking prevents unsafe prompts from ever reaching the backend.

VI

Technical Flow & Design Decisions

Design Decisions

Clean Design

Purposefully removed complex RAG layers and over-engineered orchestration to focus on speed and reliability.

Structured Output

Every response is formatted for UI rendering, ensuring explainability for the researcher.

Safe-by-Design

Validations are performed at every stage of the process, from input validation to final structured JSON output.

Workflow here

VII

Future Roadmap & Summary

Future Roadmap & Summary

Future Improvements

- Fine-tuned safety classifiers, advanced uncertainty modeling, and experiment tracking dashboards.

Summary

- NextStep Science is a safe, controlled, and explainable lab reasoning system built for the modern laboratory